

Stieltjes Antiderivatives, Hurwitz Zeta Jets, and Polygamma Functions of Negative Order

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Abstract

The generalized Stieltjes constants $\gamma_n(x)$, viewed as functions of the argument, admit a clean antiderivative calculus: with $\Gamma_n(p) := \int_1^p \gamma_n(x) dx$ one has $\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1} [\zeta^{(n+1)}(0, p) - \zeta^{(n+1)}(0)]$, and at positive integers $\Gamma_{k-1}(N+1) = -\frac{1}{k} \sum_{j=1}^N \ln^k j$ — so Γ_{k-1} is precisely the analytic interpolant of the power sums of logarithms, just as $\Gamma_0 = -\ln \Gamma$ interpolates $-\ln N!$. We work out the $s = 0$ floor of this ladder systematically: the derivative jets of $\zeta(s)$ at $s = 0$ through order five, with a conversion dictionary to the jets at $s = -1, -2$ and $s = 2, 3$; reflection formulas for $\gamma_n(p/q) - \gamma_n(1 - p/q)$ at orders $n \leq 4$, with the general mechanism at every order; closed forms for the moments $\int_0^1 x^m \zeta(s, x) dx$ and their variants; and explicit values of $\Gamma_1, \dots, \Gamma_4$ at small rational arguments, where the differences $\gamma_n(p) - \gamma_n(1 - p)$ emerge as the precise obstruction layer. A second part develops the dual ladder — the polygamma functions of negative integer order — with the same tools: reflection formulas at the $\psi^{(-n)}$ -level (filling a gap noted by Espinosa and Moll), a uniform conjugated harmonic-number bridge between the Dirichlet L -jets at $s = k+1$ and $s = -k$, a residual-dimension law for the grids $\{\zeta'(-k, j/q)\}$, and explicit values through the first denominators where real-quadratic and quartic L -derivatives enter.

1 Introduction and notation

The generalized Stieltjes constants $\gamma_n(x)$ are defined by the Laurent expansion of the Hurwitz zeta function at its pole,

$$\zeta(s, x) = \frac{1}{s-1} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(x) (s-1)^n, \quad (1)$$

so that $\gamma_0(x) = -\psi(x)$ and $\gamma_n(1) = \gamma_n$ are the classical Stieltjes constants, $\gamma_0 = \gamma$. Although universally called *constants*, the $\gamma_n(x)$ are real-analytic functions of the argument on $(0, \infty)$, and it is this function-theoretic point of view we adopt. Following Blagouchine [5, 6] we write

$$\Gamma_n(p) := \int_1^p \gamma_n(x) dx, \quad \text{so that } \Gamma_n(1) = 0,$$

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for the antiderivative normalized at 1. (The notation is Blagouchine's; the clash with the Euler Γ -function is resolved by the subscript, and indeed $\Gamma_0(p) = -\ln \Gamma(p)$, as Theorem 2.1 makes plain.) Throughout, $\zeta^{(k)}(s_0, a) := \partial_s^k \zeta(s, a)|_{s=s_0}$ and $\zeta^{(k)}(s_0) := \zeta^{(k)}(s_0, 1)$ denote derivative jets with respect to the *order*; $\Lambda_q := \gamma + \ln(2\pi q)$; $H_n = \sum_{i \leq n} 1/i$ and $H_n^{(2)} = \sum_{i \leq n} 1/i^2$ are harmonic numbers; A is the Glaisher–Kinkelin constant, $\ln A = \frac{1}{12} - \zeta'(-1)$; and

$$S_j\left(\frac{p}{q}\right) := \sum_{k=1}^{q-1} \sin \frac{2\pi kp}{q} \zeta^{(j)}\left(0, \frac{k}{q}\right), \quad S_0\left(\frac{p}{q}\right) = \frac{1}{2} \cot \frac{\pi p}{q}, \quad S_1\left(\frac{p}{q}\right) = \sum_{k=1}^{q-1} \sin \frac{2\pi kp}{q} \ln \Gamma\left(\frac{k}{q}\right)$$

(the closed forms for S_0, S_1 use $\zeta(0, a) = \frac{1}{2} - a$, $\zeta'(0, a) = \ln \Gamma(a) - \frac{1}{2} \ln 2\pi$, and $\sum_k \sin(2\pi kp/q) = 0$).

The starting observation is elementary but organizing: differentiating (1) in x against $\partial_x \zeta(s, x) = -s \zeta(s+1, x)$ identifies every derivative and antiderivative of every γ_n with a derivative jet of the Hurwitz zeta function at an *integer* value of s . Antiderivatives live at $s=0$; the functions themselves at $s=1$; their x -derivatives at $s=2$ [9] — a single ladder.

The article is organized around that calculus. Section 2 proves the ladder identity and the interpolation theorem for sums of powers of logarithms. Section 3 assembles the ζ -jets at $s=0$ through order five and the conversion dictionary (including the Glaisher-constant spelling $\zeta'(2) = \zeta(2)(\gamma + \ln 2\pi - 12 \ln A)$); the order-five jet, and the fully expanded Stieltjes-vocabulary form of the order-four jet, we have not found in the references consulted here. Section 4 derives the reflection formulas at orders up to four with the general mechanism; Section 5 the Hurwitz-transform moment evaluations; Section 6 the explicit values of Γ_1 (denominators 2, 3, 4, 6), Γ_2 and Γ_3 (at $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}$), and $\Gamma_4(\frac{1}{2})$, with the fully expanded third- and quarter-argument forms collected in the appendix. Section 7 is a second main act: the same Hurwitz-jet calculus has a dual direction, downward through the polygamma functions of negative integer order, and the reflection/multiplication/ bridge machinery transfers wholesale.

Verification protocol. Beyond proof, every displayed identity in this article has been checked numerically to at least 40 significant digits (most to 50 or more) directly from the defining Hurwitz-zeta jets; the verification scripts, with per-display residuals, live under `src/PolyLog/tools/` in the repository snapshot recorded on the title page.

2 The ladder

Theorem 2.1 (antiderivative ladder). *For $n \geq 0$ and $p > 0$,*

$$\frac{d}{dp} \zeta^{(n+1)}(0, p) = (-1)^{n+1} (n+1) \gamma_n(p), \quad (2)$$

and consequently

$$\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1} \left[\zeta^{(n+1)}(0, p) - \zeta^{(n+1)}(0) \right]. \quad (3)$$

Proof. Differentiating $\partial_x \zeta(s, x) = -s \zeta(s+1, x)$ is legitimate termwise for $\operatorname{Re} s > 1$ and extends by analytic continuation. Expand both sides around $s=0$: on the right, $-s \zeta(s+1, x) = -s \left[\frac{1}{s} + \sum_k \frac{(-1)^k}{k!} \gamma_k(x) s^k \right] = -1 - \sum_k \frac{(-1)^k}{k!} \gamma_k(x) s^{k+1}$, whose s^{n+1} -coefficient is $(-1)^{n+1} \gamma_n(x)/n!$. On the left the s^{n+1} -coefficient is $\partial_x \zeta^{(n+1)}(0, x)/(n+1)!$. Equating gives (2); integrating from 1 to p gives (3). $\square$

The differential form (2) is Lemma 2 of Chakraborty–Kanemitsu–Kuzumaki [7] (for $n = 1$ it is Deninger’s R' -law [11]); the integrated form (3) is eq. (2.16) of Connon [10]. For $n = 0$, $\zeta'(0, p) = \ln \Gamma(p) - \frac{1}{2} \ln 2\pi$ recovers $\Gamma_0(p) = -\ln \Gamma(p)$.

Theorem 2.2 (interpolation of power sums of logarithms). *For $k \geq 1$ and integer $N \geq 0$,*

$$\Gamma_{k-1}(N+1) = -\frac{1}{k} \sum_{j=1}^N \ln^k j.$$

Proof. From $\zeta(s, N+1) = \zeta(s) - \sum_{j=1}^N j^{-s}$, $\zeta^{(k)}(0, N+1) = \zeta^{(k)}(0) - \sum_j (-\ln j)^k = \zeta^{(k)}(0) - (-1)^k \sum_j \ln^k j$. Insert into (3) with $n = k-1$. $\square$

Thus Γ_{k-1} is *the* analytic interpolant of the power sums $\sum \ln^k j$ (up to the factor $-1/k$), exactly as $\ln \Gamma$ interpolates $\ln N!$: the heuristic “differentiate $1^r + \dots + N^r = \zeta(-r) - \zeta(-r, N+1)$ k times at $r = 0$ ” becomes precise through Theorem 2.1.

Corollary 2.3 (unit translates). *For $k \geq 0$ and integer $n \geq 1$,*

$$\int_n^{n+1} \gamma_k(x) dx = -\frac{\ln^{k+1} n}{k+1}; \quad \text{in particular} \quad \int_1^2 \gamma_k(x) dx = 0.$$

(The case $n = 1$ is eqs. (2.18)–(2.19) of [10].)

3 Zeta jets and the conversion dictionary

Making (3) explicit requires the constants $\zeta^{(k)}(0)$ and, for the moment evaluations of Section 5, the jets at negative integers. All of them follow mechanically from the functional equation $\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$ by expanding $\Gamma(1-s) = \exp(\gamma s + \sum_{k \geq 2} \zeta(k) s^k / k)$ and the Laurent series of $\zeta(1-s)$ at the pole.

Proposition 3.1 (jets at 0). *With $L := \ln 2\pi$,*

$$\begin{aligned} \zeta'(0) &= -\frac{L}{2}, \\ \zeta''(0) &= \frac{\gamma^2}{2} + \gamma_1 - \frac{\pi^2}{24} - \frac{L^2}{2}, \\ \zeta'''(0) &= \gamma^3 + 3\gamma\gamma_1 + \frac{3}{2}\gamma_2 - \zeta(3) + \frac{3}{2}\gamma^2 L + 3\gamma_1 L - \frac{\pi^2}{8} L - \frac{L^3}{2}, \\ \zeta^{(4)}(0) &= \frac{3}{2}\gamma^4 + 4\gamma^3 L + 6\gamma^2 \gamma_1 + 3\gamma^2 L^2 + \frac{\pi^2}{4}\gamma^2 + 12\gamma\gamma_1 L + 6\gamma\gamma_2 + 6\gamma_1 L^2 + \frac{\pi^2}{2}\gamma_1 \\ &\quad + 6\gamma_2 L + 2\gamma_3 - \frac{L^4}{2} - \frac{\pi^2}{4} L^2 - 4\zeta(3)L - \frac{19\pi^4}{480}. \end{aligned}$$

Proposition 3.2 (fifth jet at 0). *With $L := \ln 2\pi$,*

$$\begin{aligned} \zeta^{(5)}(0) &= 2\gamma^5 + \frac{15}{2}\gamma^4 L + \gamma^3 \left(10L^2 + \frac{5\pi^2}{6}\right) + \gamma^2 \left(5L^3 + \frac{5\pi^2}{4} L + 10\zeta(3)\right) \\ &\quad + \gamma\gamma_1 \left(30L^2 + \frac{5\pi^2}{2}\right) + \gamma_1 \left(10\gamma^3 + 30\gamma^2 L + 10L^3 + \frac{5\pi^2}{2} L + 20\zeta(3)\right) \\ &\quad + \gamma_2 \left(15\gamma^2 + 30\gamma L + 15L^2 + \frac{5\pi^2}{4}\right) + \gamma_3 (10\gamma + 10L) + \frac{5}{2}\gamma_4 \\ &\quad - \zeta(3) \left(10L^2 + \frac{5\pi^2}{6}\right) - 12\zeta(5) - \frac{L^5}{2} - \frac{5\pi^2}{12} L^3 - \frac{19\pi^4}{96} L. \end{aligned}$$

The values $\zeta''(0)$ and $\zeta'''(0)$ are classical (Ramanujan [4]; Apostol [2]; Choudhury [8]). Apostol's Theorem 3 [2] in fact gives a closed form of $\zeta^{(n)}(0)$ at every order, in terms of the Laurent coefficients of $\Gamma(s)\zeta(s)$ at $s = 1$, and prints the example $n = 4$ in that vocabulary; we have not located the fully expanded forms above — nor any explicit form of $\zeta^{(5)}(0)$ — in print. The pattern continues to every order by the same expansion, which packages into the following closed recurrence.

Remark 3.3 (explicit recurrence for the jets at 0). In $\zeta(s) = \chi(s)\zeta(1-s)$ with $\chi(s) = 2^s\pi^{s-1}\sin(\pi s/2)\Gamma(1-s)$, the simple zero of χ at $s = 0$ cancels the pole of $\zeta(1-s)$; factor it out as $\chi(s) = \frac{s}{2}e^{g(s)}$ with $g(s) := \ln(2\chi(s)/s) = \ln[(2\pi)^s \frac{\sin(\pi s/2)}{\pi s/2} \Gamma(1-s)]$, analytic on $|s| < 1$ and real on $(-1, 1)$ (the bracket is positive there), $g(0) = 0$. From $\ln \Gamma(1-s) = \gamma s + \sum_{k \geq 2} \zeta(k)s^k/k$ and $\ln[\sin(\pi s/2)/(\pi s/2)] = -\sum_{m \geq 1} \zeta(2m)(s/2)^{2m}/m$,

$$g(s) = (\gamma + L)s + \sum_{k \geq 2} c_k s^k, \quad c_k = \frac{\zeta(k)}{k} \cdot \begin{cases} 1, & k \text{ odd}, \\ 1 - 2^{1-k}, & k \text{ even}, \end{cases}$$

so $c_2 = \frac{\pi^2}{24}$, $c_3 = \frac{\zeta(3)}{3}$, $c_4 = \frac{7\pi^4}{2880}$, $c_5 = \frac{\zeta(5)}{5}$. Since $s\zeta(1-s) = -1 + \sum_{m \geq 1} \gamma_{m-1} s^m/(m-1)!$ is entire, the Leibniz rule gives

$$\zeta^{(n)}(0) = \frac{1}{2} \sum_{k=0}^n \binom{n}{k} B_k(1!c_1, 2!c_2, \dots, k!c_k) h_{n-k}, \quad h_0 = -1, \quad h_m = m\gamma_{m-1} \ (m \geq 1),$$

where $c_1 = \gamma + L$ and B_k is the complete Bell polynomial, $B_k(g'(0), \dots, g^{(k)}(0)) = (e^g)^{(k)}(0)$: $B_0 = 1$, $B_1 = c_1$, $B_2 = c_1^2 + 2c_2$, $B_3 = c_1^3 + 6c_1c_2 + 6c_3$. The cases $n = 0, 1$ recover $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{L}{2}$; $n = 2, \dots, 5$ reproduce the two propositions verbatim.

Proposition 3.4 (jets at -1 and -2). *Expanding the functional equation around $s = -1$ (where $\zeta(2-\varepsilon)$ is regular) and $s = -2$ (a trivial zero) gives, with $L = \ln 2\pi$:*

$$\zeta(-1) = -\frac{1}{12}, \quad \zeta'(-1) = \frac{\zeta'(2)}{2\pi^2} + \frac{1-\gamma-L}{12}, \quad (4)$$

$$\zeta''(-1) = -\frac{\zeta''(2)}{2\pi^2} + \frac{\zeta'(2)}{\pi^2}(\gamma+L-1) - \frac{(\gamma+L)^2}{12} + \frac{\gamma+L}{6} + \frac{\pi^2}{144}, \quad (5)$$

$$\zeta(-2) = 0, \quad \zeta'(-2) = -\frac{\zeta(3)}{4\pi^2}, \quad \zeta''(-2) = \frac{\zeta(3)}{4\pi^2}(3-2\gamma-2L) + \frac{\zeta'(3)}{2\pi^2}. \quad (6)$$

Remark 3.5 (Glaisher spelling). Since $\ln A = \frac{1}{12} - \zeta'(-1)$, eq. (4) is equivalent to

$$\zeta'(2) = \zeta(2)(\gamma + \ln 2 + \ln \pi - 12 \ln A),$$

the form returned by Mathematica's `FunctionExpand[Zeta'[2]]`. Either spelling may be used in all the moment formulas below; we keep $\zeta'(2), \zeta''(2), \zeta'(3)$ as the canonical second-layer constants.

4 Reflection formulas

Hurwitz's formula (see, e.g., [3, Ch. 12]), for $0 < p/q < 1$,

$$\zeta\left(1-s, \frac{p}{q}\right) = \frac{2\Gamma(s)}{(2\pi q)^s} \sum_{k=1}^q \left[\cos\left(\frac{\pi s}{2} - \frac{2\pi kp}{q}\right) \right] \zeta\left(s, \frac{k}{q}\right),$$

expanded around $s = 0$, expresses every $\gamma_n(p/q)$ through the jets $\zeta^{(j)}(0, k/q)$, $j \leq n + 1$. Taking the part odd under $p \mapsto q - p$ isolates the sine sums S_j and yields reflection formulas. Write $E(s) := \exp(-\Lambda_q s + \sum_{k \geq 2} (-1)^k \zeta(k) s^k/k) = \sum_i e_i s^i$ for the prefactor expansion $(2\Gamma(s)(2\pi q)^{-s} = \frac{2}{s} E(s))$.

Theorem 4.1 (reflection, general mechanism). *For $n \geq 1$ and $0 < p < q$,*

$$\gamma_n\left(\frac{p}{q}\right) - \gamma_n\left(1 - \frac{p}{q}\right) = 4n! \sum_{i+2l+1+j=n+1} e_i \frac{(-1)^l (\pi/2)^{2l+1}}{(2l+1)!} \frac{S_j(p/q)}{j!}.$$

Explicitly, with $\Lambda = \Lambda_q = \gamma + \ln 2\pi q$ (the letter distinguishes it from the Glaisher constant Λ):

$$\gamma_2\left(\frac{p}{q}\right) - \gamma_2\left(1 - \frac{p}{q}\right) = 2\pi S_2 - 4\pi\Lambda S_1 + \left(2\Lambda^2 + \frac{\pi^2}{6}\right)\pi S_0, \quad (7)$$

$$\gamma_3\left(\frac{p}{q}\right) - \gamma_3\left(1 - \frac{p}{q}\right) = 2\pi S_3 - 6\pi\Lambda S_2 + \left(6\Lambda^2 + \frac{\pi^2}{2}\right)\pi S_1 - \left(2\Lambda^3 + \frac{\pi^2}{2}\Lambda + 4\zeta(3)\right)\pi S_0, \quad (8)$$

$$\begin{aligned} \gamma_4\left(\frac{p}{q}\right) - \gamma_4\left(1 - \frac{p}{q}\right) &= 2\pi S_4 - 8\pi\Lambda S_3 + (12\Lambda^2 + \pi^2)\pi S_2 - (8\Lambda^3 + 2\pi^2\Lambda + 16\zeta(3))\pi S_1 \\ &\quad + \left(2\Lambda^4 + \pi^2\Lambda^2 + 16\Lambda\zeta(3) + \frac{19\pi^4}{120}\right)\pi S_0. \end{aligned} \quad (9)$$

Proof. Split the cosine, $\cos(\pi s/2 - \theta) = \cos(\pi s/2)\cos\theta + \sin(\pi s/2)\sin\theta$. Under $p \mapsto q - p$ the cosine sums are even and the sine sums odd, so the odd part of $\zeta(1 - s, p/q)$ equals $\frac{2}{s} E(s) \sin(\pi s/2) \sum_j S_j s^j/j!$. By (1), $\zeta(1 - s, p/q) = -\frac{1}{s} + \sum_n \gamma_n(p/q) s^n/n!$, whose odd part has s^n -coefficient $\frac{1}{2n!} [\gamma_n(p/q) - \gamma_n(1 - p/q)]$. Matching coefficients of s^n (i.e. s^{n+1} of $E \cdot \sin \cdot \sum S_j s^j/j!$) gives the general formula; instantiating $e_0 = 1$, $e_1 = -\Lambda$, $e_2 = \frac{\Lambda^2}{2} + \frac{\pi^2}{12}$, $e_3 = -\frac{\Lambda^3}{6} - \frac{\pi^2}{12}\Lambda - \frac{\zeta(3)}{3}$, $e_4 = \frac{\Lambda^4}{24} + \frac{\pi^2}{24}\Lambda^2 + \frac{\zeta(3)}{3}\Lambda + \frac{\pi^4}{288} + \frac{\pi^4}{360}$ yields (7)–(9). $\square$

Formula (7) is eq. (89) of Blagouchine [6], where a footnote records that it also appears in unpublished work of D. Cannon; (8) and (9) appear to be new. The same expansion, one order down, recovers Malmsten's reflection formula for γ_1 [6]; an elementary alternative proof of that $n = 1$ case, via Kummer's Fourier expansion of $\ln \Gamma$ rather than Hurwitz's functional equation, is given by Kouba [15]. For $q = 3$ and $q = 4$ the sums collapse ($S_j(\frac{1}{3}) = \frac{\sqrt{3}}{2} [\zeta^{(j)}(0, \frac{1}{3}) - \zeta^{(j)}(0, \frac{2}{3})]$, $S_j(\frac{1}{4}) = \zeta^{(j)}(0, \frac{1}{4}) - \zeta^{(j)}(0, \frac{3}{4})$), so (7)–(9) can be *inverted* to express the odd jet differences through the Stieltjes differences $\gamma_n(p) - \gamma_n(1 - p)$ — the key to Section 6.

5 Moments

Theorem 5.1 (Hurwitz transform and variants). *For $m \geq 1$ and $\operatorname{Re} s < 1$,*

$$\int_0^1 x^m \zeta(s, x) dx = - \sum_{j=1}^m \frac{m!}{(m-j+1)!} \frac{\zeta(s-j)}{(s-1)(s-2)\cdots(s-j)}, \quad (10)$$

and $\int_0^1 \zeta(s, x) dx = 0$. The same recursion with boundary terms $\zeta(\sigma, \frac{1}{2}) = (2^\sigma - 1)\zeta(\sigma)$ and $\zeta(\sigma, 2) = \zeta(\sigma) - 1$ gives the analogous closed forms for $\int_{1/2}^1$ and $\int_1^2$.

Proof. Since $\zeta(s, x) = -\partial_x \zeta(s-1, x)/(s-1)$, integration by parts gives $I_m(s) = -\zeta(s-1)/(s-1) + \frac{m}{s-1} I_{m-1}(s-1)$ with $I_0 \equiv 0$ (the boundary term at $x = 0$ vanishes for $\operatorname{Re} s < m+1$ since $\zeta(\sigma, x) = x^{-\sigma} + O(1)$); unrolling yields (10), which then extends meromorphically in s . $\square$

For the unit interval, (10) is Theorem 3.7 of Espinosa–Moll [13]. Expanding (10) at $s = 1 + t$ against $\int_0^1 x^m \zeta(1 + t, x) dx = \frac{1}{(m+1)t} + M_0 - M_1 t + \frac{M_2}{2} t^2 - \dots$, where $M_k = \int_0^1 x^m \gamma_k(x) dx$, reads off every Stieltjes moment. The expansion $1/[(t-1) \cdots (t-j+1)] = \frac{(-1)^{j-1}}{(j-1)!} (1 + H_{j-1} t + \frac{H_{j-1}^2 + H_{j-1}^{(2)}}{2} t^2 + \dots)$ gives the coefficients in closed form:

Theorem 5.2 (general first moment). *For every integer $m \geq 1$,*

$$\int_0^1 x^m \gamma_1(x) dx = \sum_{j=1}^m (-1)^{j-1} \binom{m}{j-1} \left[\frac{\zeta''(1-j)}{2} + H_{j-1} \zeta'(1-j) + \frac{H_{j-1}^2 + H_{j-1}^{(2)}}{2} \zeta(1-j) \right].$$

The γ_n -moment carries the analogous formula with the order- $(n+1)$ jet leading and complete homogeneous symmetric functions of $1, \frac{1}{2}, \dots, \frac{1}{j-1}$ as coefficients.

Combining Theorems 5.1 and 5.2 with the dictionary of Proposition 3.4 produces the following table. Items (11)–(14) are restated from the corpus that motivated this work; the remaining evaluations appear to be new.

$$\int_{1/2}^1 \gamma_1(x) dx = \frac{\gamma_1}{2} + \frac{\gamma^2}{4} - \frac{\pi^2}{48} + \frac{\ln^2 2}{2} - \frac{\ln^2 \pi}{4}, \quad (11)$$

$$\int_1^2 x \gamma_1(x) dx = 1 + \frac{\gamma_1}{2} + \frac{\gamma^2}{4} - \frac{\pi^2}{48} - \frac{\ln^2 2\pi}{4}, \quad (12)$$

$$\int_0^1 x^2 \gamma_1(x) dx = \frac{\gamma_1}{2} + \frac{\gamma^2}{3} - \frac{\pi^2}{36} - \frac{\ln^2 2\pi}{6} + \frac{\gamma}{6} \ln 2\pi - \frac{\zeta'(2)}{\pi^2} (\gamma + \ln 2\pi) + \frac{\zeta''(2)}{2\pi^2}, \quad (13)$$

$$\begin{aligned} \int_{1/2}^1 x \gamma_1(x) dx &= \frac{\gamma_1}{2} + \frac{5\gamma^2}{16} - \frac{5\pi^2}{192} + \frac{\ln^2 2}{8} - \frac{\ln 2 \ln \pi}{6} - \frac{3 \ln^2 \pi}{16} + \gamma \left(\frac{\ln 2}{12} + \frac{\ln \pi}{8} \right) \\ &\quad - \frac{\zeta'(2)}{4\pi^2} (3\gamma + 2 \ln 2 + 3 \ln \pi) + \frac{3\zeta''(2)}{8\pi^2}, \end{aligned} \quad (14)$$

$$\int_0^1 x \gamma_1(x) dx = \frac{\zeta''(0)}{2} = \frac{\gamma^2}{4} + \frac{\gamma_1}{2} - \frac{\pi^2}{48} - \frac{\ln^2 2\pi}{4}, \quad (15)$$

$$\begin{aligned} \int_0^1 x^3 \gamma_1(x) dx &= \frac{3\gamma^2}{8} + \frac{\gamma_1}{2} - \frac{\pi^2}{32} + \frac{\gamma \ln 2\pi}{4} - \frac{\ln^2 2\pi}{8} - \frac{3(\zeta(3) + 2\zeta'(2))}{4\pi^2} (\gamma + \ln 2\pi) \\ &\quad + \frac{3(\zeta'(3) + \zeta''(2))}{4\pi^2}, \end{aligned} \quad (16)$$

$$\int_1^2 x^2 \gamma_1(x) dx = \frac{9}{4} + \frac{3\gamma_1}{2} + \frac{5\gamma^2}{6} - \frac{5\pi^2}{72} + \frac{\gamma \ln 2\pi}{6} - \frac{2 \ln^2 2\pi}{3} - \frac{\zeta'(2)}{\pi^2} (\gamma + \ln 2\pi) + \frac{\zeta''(2)}{2\pi^2}, \quad (17)$$

$$\int_0^1 x \gamma_2(x) dx = -\frac{\zeta'''(0)}{3} = \frac{\zeta(3)}{3} - \frac{\gamma^3}{3} - \gamma\gamma_1 - \frac{\gamma_2}{2} - \frac{\gamma^2}{2} \ln 2\pi - \gamma_1 \ln 2\pi + \frac{\pi^2}{24} \ln 2\pi + \frac{\ln^3 2\pi}{6}, \quad (18)$$

$$\int_1^2 x \gamma_2(x) dx = \int_0^1 x \gamma_2(x) dx - 2, \quad \int_{1/2}^1 \gamma_2(x) dx = -\Gamma_2\left(\frac{1}{2}\right). \quad (19)$$

At $m = 3$ the constants $\zeta(3)$ and $\zeta'(3)$ enter through (6), and each further unit of m opens one more column of jets at the next negative integer; the second moment of γ_1 over $[1/2, 1]$ and the higher γ_2 -moments are equally mechanical but bulkier, and we omit them.

6 Explicit values of Γ_n

6.1 The first antiderivative

Theorem 6.1 (Γ_1 in closed form). *For all $p > 0$,*

$$\Gamma_1(p) = c + \frac{1}{2} \zeta''(0, p), \quad c := -\frac{1}{2} \zeta''(0) = \frac{\pi^2}{48} + \frac{\ln^2 2\pi}{4} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2}.$$

Moreover $\int_0^1 \Gamma_1(x) dx = c$, equivalently (15) (since $\int_0^1 \zeta''(0, x) dx = 0$ [11, 9]).

Theorem 6.2 (half-integral and even-sum values).

$$\begin{aligned} \Gamma_1\left(\frac{1}{2}\right) &= \frac{\pi^2}{48} - \frac{\gamma^2}{4} - \frac{\gamma_1}{2} - \frac{\ln^2 2}{4} - \frac{\ln 2 \ln 2\pi}{2} + \frac{\ln^2 2\pi}{4}, \\ \Gamma_1\left(\frac{1}{3}\right) + \Gamma_1\left(\frac{2}{3}\right) &= 2c - \frac{\ln^2 3}{4} - \frac{\ln 3 \ln 2\pi}{2}, \quad \Gamma_1\left(\frac{1}{4}\right) + \Gamma_1\left(\frac{3}{4}\right) = 2c - \frac{3 \ln^2 2}{4} - \frac{\ln 2 \ln 2\pi}{2}, \\ \Gamma_1\left(\frac{1}{6}\right) + \Gamma_1\left(\frac{5}{6}\right) &= 2c - \frac{\ln 2 \ln 3}{2}. \end{aligned}$$

Proof. $\zeta(s, \frac{1}{2}) = (2^s - 1)\zeta(s)$ gives $\zeta''(0, \frac{1}{2}) = -\frac{1}{2} \ln^2 2 - \ln 2 \ln 2\pi$; the multiplication identities $\zeta(s, \frac{1}{3}) + \zeta(s, \frac{2}{3}) = (3^s - 1)\zeta(s)$, $\zeta(s, \frac{1}{4}) + \zeta(s, \frac{3}{4}) = (4^s - 2^s)\zeta(s)$, $\zeta(s, \frac{1}{6}) + \zeta(s, \frac{5}{6}) = (6^s - 3^s - 2^s + 1)\zeta(s)$, twice differentiated at $s = 0$, give the even sums (for $q = 6$ the $\zeta''(0)$ -coefficient $\ln^2 6 - \ln^2 3 - \ln^2 2 = 2 \ln 2 \ln 3$ and the $\zeta'(0)$ -coefficient $\ln 6 - \ln 3 - \ln 2 = 0$ produce the strikingly short third identity). Apply Theorem 6.1. $\square$

Theorem 6.3 (third and quarter values).

$$\begin{aligned} \Gamma_1\left(\frac{1}{3}\right) &= \frac{\pi^2}{72} - \frac{\ln^2 2\pi}{3} + \frac{\ln^2 3}{24} - \frac{2}{3} \ln 3 \ln 2\pi - \frac{\gamma}{12} (4\gamma + 8 \ln 2\pi - \ln 3) + (\gamma + \ln 6\pi) \ln \Gamma\left(\frac{1}{3}\right) \\ &\quad - \frac{\gamma_1}{2} + \frac{\sqrt{3}}{12\pi} \left(\gamma_2\left(\frac{1}{3}\right) - \gamma_2\left(\frac{2}{3}\right) \right), \\ \Gamma_1\left(\frac{1}{4}\right) &= \frac{\pi^2}{96} - \frac{3}{8} \ln^2 \pi - \frac{9}{4} \ln 2 \ln 2\pi - \frac{\gamma}{8} (3\gamma + 8 \ln 2 + 6 \ln \pi) + (\gamma + \ln 8\pi) \ln \Gamma\left(\frac{1}{4}\right) \\ &\quad - \frac{\gamma_1}{2} + \frac{1}{8\pi} \left(\gamma_2\left(\frac{1}{4}\right) - \gamma_2\left(\frac{3}{4}\right) \right), \end{aligned}$$

and $\Gamma_1(\frac{2}{3})$, $\Gamma_1(\frac{3}{4})$ follow from Theorem 6.2. Furthermore, with the abbreviation $D_j := \zeta^{(j)}(0, \frac{1}{3}) - \zeta^{(j)}(0, \frac{2}{3})$,

$$\Gamma_1\left(\frac{1}{6}\right) - \Gamma_1\left(\frac{5}{6}\right) = \frac{\ln 2 \ln 18}{6} + \ln 2 \left(2 \ln \Gamma\left(\frac{1}{3}\right) + \frac{\ln 3}{6} - \ln 2 - \ln \pi \right) + D_2,$$

which together with the even sum of Theorem 6.2 gives $\Gamma_1(\frac{1}{6})$ and $\Gamma_1(\frac{5}{6})$ explicitly — in the same obstruction atom $\gamma_2(\frac{1}{3}) - \gamma_2(\frac{2}{3})$, since inverting (7) at $q = 3$ expresses $D_2 = \frac{2}{\sqrt{3}} S_2(\frac{1}{3})$ through it.

Proof. By Theorem 6.1 everything reduces to the single jets $\zeta''(0, p)$ at $p = \frac{1}{3}, \frac{1}{4}, \frac{1}{6}$. Their even parts are in Theorem 6.2; their odd parts are the inversions of (7) at $q = 3, 4$:

$$S_2 = \frac{1}{2\pi} \left[\gamma_2\left(\frac{p}{q}\right) - \gamma_2\left(1 - \frac{p}{q}\right) + 4\pi\Lambda S_1 - \left(2\Lambda^2 + \frac{\pi^2}{6}\right) \pi S_0 \right],$$

with $S_1(\frac{1}{3}) = \frac{\sqrt{3}}{2}(2\ln\Gamma(\frac{1}{3}) - \ln\frac{2\pi}{\sqrt{3}})$, $S_0(\frac{1}{3}) = \frac{1}{2\sqrt{3}}$, $S_1(\frac{1}{4}) = 2\ln\Gamma(\frac{1}{4}) - \ln(\pi\sqrt{2})$, $S_0(\frac{1}{4}) = \frac{1}{2}$. For $q = 6$, the Dirichlet character χ_{-3} gives $\zeta(s, \frac{1}{6}) - \zeta(s, \frac{5}{6}) = (6^s + 3^s)L(s, \chi_{-3})$ while $\zeta(s, \frac{1}{3}) - \zeta(s, \frac{2}{3}) = 3^s L(s, \chi_{-3})$; eliminating the jets of $L(s, \chi_{-3})$ at $s = 0$ ($L(0, \chi_{-3}) = \frac{1}{3}$, $L'(0, \chi_{-3}) = 2\ln\Gamma(\frac{1}{3}) + \frac{\ln 3}{6} - \ln 2 - \ln \pi$) yields the displayed transport identity. $\square$

6.2 The second and third antiderivatives

Theorem 6.4 (Γ_2 values).

$$\Gamma_2(\frac{1}{2}) = \frac{\gamma_2}{2} + \frac{\gamma^3}{3} - \frac{\zeta(3)}{3} + \ln^3 2 - \frac{\ln^3 \pi}{6} + \frac{\ln \pi}{24} (12\gamma^2 - \pi^2 + 24\ln^2 2) + (\gamma + \ln \pi) \gamma_1,$$

and at the third and quarter points, with $Z_3(\frac{1}{q}) := \zeta'''(0, \frac{1}{q}) + \zeta'''(0, 1 - \frac{1}{q})$ and $D_3(\frac{1}{q}) := \zeta'''(0, \frac{1}{q}) - \zeta'''(0, 1 - \frac{1}{q})$,

$$\Gamma_2(\frac{1}{q}) = -\frac{1}{3} \left[\frac{Z_3 + D_3}{2} - \zeta'''(0) \right], \quad q \in \{3, 4\},$$

where the even parts are elementary,

$$Z_3(\frac{1}{3}) = -\frac{\ln^3 3}{2} - \frac{3\ln^2 3 \ln 2\pi}{2} + 3\ln 3 \zeta''(0), \quad Z_3(\frac{1}{4}) = \partial_s^3 [(4^s - 2^s)\zeta(s)]_{s=0},$$

and the odd parts are the inversions of (8): $D_3 = \frac{2}{\sqrt{3}}S_3$ at $q = 3$, $D_3 = S_3$ at $q = 4$, with

$$S_3 = \frac{1}{2\pi} \left[\gamma_3(\frac{1}{q}) - \gamma_3(1 - \frac{1}{q}) + 6\pi\Lambda S_2 - \left(6\Lambda^2 + \frac{\pi^2}{2}\right)\pi S_1 + \left(2\Lambda^3 + \frac{\pi^2}{2}\Lambda + 4\zeta(3)\right)\pi S_0 \right]$$

and S_2 as in the proof of Theorem 6.3. Thus $\Gamma_2(\frac{1}{3})$ and $\Gamma_2(\frac{1}{4})$ are explicit in the constants $\{\gamma, \gamma_1, \gamma_2, \zeta(3), \ln 2, \ln 3, \ln \pi, \ln \Gamma(\frac{1}{q}), \gamma_2(\frac{1}{q}) - \gamma_2(1 - \frac{1}{q}), \gamma_3(\frac{1}{q}) - \gamma_3(1 - \frac{1}{q})\}$, the two Stieltjes differences being the only non-classical ingredients.

Proof. $\Gamma_2(p) = -\frac{1}{3}[\zeta'''(0, p) - \zeta'''(0)]$ by Theorem 2.1. At $p = \frac{1}{2}$ use $\zeta'''(0, \frac{1}{2}) = \partial_s^3 [(2^s - 1)\zeta(s)]_{s=0} = \ln^3 2 \zeta(0) + 3\ln^2 2 \zeta'(0) + 3\ln 2 \zeta''(0)$ and Proposition 3.1; at $p = \frac{1}{3}, \frac{1}{4}$ split into even and odd parts as displayed. $\square$

Corollary 6.5 (Γ_2 even sums). With $L = \ln 2\pi$,

$$\begin{aligned} \Gamma_2(\frac{1}{3}) + \Gamma_2(\frac{2}{3}) &= \frac{2}{3} \zeta'''(0) - \ln 3 \zeta''(0) + \frac{L}{2} \ln^2 3 + \frac{\ln^3 3}{6}, \\ \Gamma_2(\frac{1}{4}) + \Gamma_2(\frac{3}{4}) &= \frac{2}{3} \zeta'''(0) - \ln 2 \zeta''(0) + \frac{3L}{2} \ln^2 2 + \frac{7\ln^3 2}{6}, \\ \Gamma_2(\frac{1}{6}) + \Gamma_2(\frac{5}{6}) &= \frac{2}{3} \zeta'''(0) + L \ln 2 \ln 3 + \frac{\ln 2 \ln 3 \ln 6}{2}, \end{aligned}$$

so that $\Gamma_2(\frac{2}{3}), \Gamma_2(\frac{3}{4})$ follow from Theorem 6.4; as in Theorem 6.2, the $\zeta''(0)$ -term vanishes at $q = 6$ because $\ln 6 - \ln 3 - \ln 2 = 0$. (Generally, with the inclusion-exclusion combination $\zeta(s, \frac{1}{q}) + \zeta(s, 1 - \frac{1}{q}) = (\sum_i c_i m_i^s) \zeta(s)$, the even sum equals $\frac{2}{3} \zeta'''(0) + \frac{1}{6} \sum_i c_i \ln^3 m_i + \frac{L}{2} \sum_i c_i \ln^2 m_i - \zeta''(0) \sum_i c_i \ln m_i$.)

Theorem 6.6 (Γ_3 values).

$$\begin{aligned}\Gamma_3\left(\frac{1}{2}\right) = & -\frac{7\ln^4 2}{4} + \frac{\ln^4 \pi}{8} - \frac{3\gamma^4}{8} - \frac{\gamma_3}{2} + \frac{19\pi^4}{1920} + \gamma_1\left(3\ln^2 2 - \frac{3\gamma^2}{2} - \frac{3\ln^2 \pi}{2} - \frac{\pi^2}{8}\right) \\ & - \frac{3\gamma_2}{2}(\gamma + \ln \pi) + \zeta(3)\ln \pi + \gamma^2\left(\frac{3\ln^2 2}{2} - \frac{3\ln^2 \pi}{4} - \frac{\pi^2}{16}\right) - \gamma^3 \ln \pi - 3\gamma\gamma_1 \ln \pi \\ & - 3\ln \pi \ln^3 2 - \frac{3\ln^2 2 \ln^2 \pi}{2} - \frac{\pi^2 \ln^2 2}{8} + \frac{\pi^2 \ln^2 \pi}{16},\end{aligned}$$

and $\Gamma_3(\frac{1}{3}), \Gamma_3(\frac{1}{4})$ are given by $\Gamma_3(\frac{1}{q}) = \frac{1}{4}[\zeta^{(4)}(0, \frac{1}{q}) - \zeta^{(4)}(0)]$ with the even part $\partial_s^4[(q\text{-orbit}) \zeta(s)]_{s=0}$ as in Theorem 6.4 and the odd part inverted from (9); the resulting expressions are explicit in the same vocabulary augmented by $\gamma_4(\frac{1}{q}) - \gamma_4(1 - \frac{1}{q})$.

6.3 The fourth antiderivative at $1/2$

Theorem 6.7 ($\Gamma_4(1/2)$).

$$\begin{aligned}\Gamma_4\left(\frac{1}{2}\right) = & \frac{\gamma_4}{2} + 3\ln^5 2 - \frac{12\zeta(5)}{5} - \frac{\ln^5 \pi}{10} + \frac{2\gamma^5}{5} + \frac{3\gamma^4 \ln \pi}{2} \\ & + \gamma_1\left(2\gamma^3 - 12\ln^3 2 + 2\ln^3 \pi + 4\zeta(3) + \frac{\pi^2 \ln \pi}{2} - 12\ln \pi \ln^2 2 + 6\gamma^2 \ln \pi\right) \\ & + \gamma\gamma_1\left(\frac{\pi^2}{2} - 12\ln^2 2 + 6\ln^2 \pi\right) + \gamma_2\left(3\gamma^2 - 6\ln^2 2 + 3\ln^2 \pi + \frac{\pi^2}{4} + 6\gamma \ln \pi\right) \\ & + 2\gamma_3(\gamma + \ln \pi) + \zeta(3)\left(4\ln^2 2 - 2\ln^2 \pi - \frac{\pi^2}{6}\right) \\ & + \gamma^2\left(\ln^3 \pi - 6\ln^3 2 + 2\zeta(3) - 6\ln \pi \ln^2 2 + \frac{\pi^2 \ln \pi}{4}\right) + \gamma^3\left(2\ln^2 \pi - 4\ln^2 2 + \frac{\pi^2}{6}\right) \\ & + \frac{\pi^2 \ln^3 2}{2} + 2\ln^2 2 \ln^3 \pi + 6\ln^3 2 \ln^2 \pi + 7\ln \pi \ln^4 2 \\ & - \frac{19\pi^4 \ln \pi}{480} - \frac{\pi^2 \ln^3 \pi}{12} + \frac{\pi^2 \ln \pi \ln^2 2}{2}.\end{aligned}$$

Proof. $\Gamma_4(\frac{1}{2}) = -\frac{1}{5}[\zeta^{(5)}(0, \frac{1}{2}) - \zeta^{(5)}(0)]$, with

$$\zeta^{(5)}\left(0, \frac{1}{2}\right) = \partial_s^5[(2^s - 1)\zeta(s)]_{s=0} = \sum_{k=0}^4 \binom{5}{k} \ln^{5-k} 2 \zeta^{(k)}(0)$$

(the $k = 5$ term carries the factor $2^0 - 1 = 0$) and Proposition 3.2. $\square$

The pattern of Theorems 6.4–6.7 is general: each rung Γ_n adds one Stieltjes constant γ_n , one odd zeta value, and one degree to the logarithmic monomials at $p = \frac{1}{2}$; at $p = \frac{1}{3}, \frac{1}{4}$ it adds the single new obstruction atom $\gamma_{n+1}(p) - \gamma_{n+1}(1 - p)$, delivered by the order- $(n + 1)$ reflection formula of Theorem 4.1. For denominators $q \geq 5$ the even parts cease to be elementary (the multiplication identities no longer close over the principal character), and the values $\Gamma_n(p/q)$ additionally involve the even-character combinations of $\zeta^{(n+1)}(0, \cdot)$ — the obstruction layer studied, at $n = 1$, in [6].

7 The negative-order polygamma floor

The ladder of Section 2 integrates in the *Laurent index* at $s = 1$. The same Hurwitz-jet calculus has a second, equally natural direction, and this section — a second main act rather than an appendix — follows it downward. Integrating in the *argument* at the s -jet floor produces the polygamma functions of negative integer order ("negapolygamma", a term Adamchik [1] credits to Gosper): for real $z > 0$,

$$\psi^{(-1)}(z) = \ln \Gamma(z), \quad \psi^{(-n)}(z) = \int_0^z \psi^{(-(n-1))}(t) dt = \frac{1}{(n-2)!} \int_0^z (z-t)^{n-2} \ln \Gamma(t) dt,$$

which is precisely Mathematica's `PolyGamma[-n, z]` (all identities below are stated on $0 < z < 1$, where the rational arguments live, and extend to $z > 0$ by the same formulas wherever both sides are defined). Everything built for the Stieltjes side transfers wholesale: the ladder identity has a one-line analogue, the reflection mechanism of Section 4 reappears at the jet floor with bare Clausen values in place of the sine sums S_j , the obstruction constants again organize along Dirichlet characters — and the bridge identities for those constants close a circle the Stieltjes side leaves open. The tools: a one-line ladder, reflection and multiplication laws, bridge identities, and explicit rational-argument values.

7.1 The ladder

Theorem 7.1. *For $n \geq 2$ and $z > 0$,*

$$\psi^{(-n)}(z) = \frac{\zeta'(1-n, z) - \zeta'(1-n)}{(n-1)!} + Q_n(z),$$

where Q_n is an explicit polynomial with coefficients in $\mathbb{Q}[\ln 2\pi, \zeta'(-1), \dots, \zeta'(2-n)]$:

$$\begin{aligned} Q_2(z) &= \frac{z}{2} \ln 2\pi + \frac{z(1-z)}{2}, \\ Q_3(z) &= \frac{z^2}{4} \ln 2\pi - \frac{z^3}{4} + \frac{3z^2}{8} - \frac{z}{24} - z \zeta'(-1), \\ Q_4(z) &= \frac{z^3}{12} \ln 2\pi - \frac{11z^4}{144} + \frac{11z^3}{72} - \frac{5z^2}{144} - \frac{z^2}{2} \zeta'(-1) - \frac{z}{2} \zeta'(-2), \end{aligned}$$

and in general Q_n has leading term $-H_{n-1}z^n/n!$ and $\ln 2\pi$ -coefficient $z^{n-1}/(2(n-1)!)$.

Proof. $\int_0^z \zeta(s, t) dt = -[\zeta(s-1, z) - \zeta(s-1)]/(s-1)$ for $\operatorname{Re} s < 1$; differentiating at $s = -k$,

$$\int_0^z \zeta'(-k, t) dt = \frac{\zeta'(-k-1, z) - \zeta'(-k-1)}{k+1} + \frac{\zeta(-k-1, z) - \zeta(-k-1)}{(k+1)^2},$$

with $\zeta(-m, z) = -B_{m+1}(z)/(m+1)$. Iterate from $\psi^{(-1)} = \zeta'(0, z) + \frac{1}{2} \ln 2\pi$: each step shifts the single ζ' -atom by one and deposits Bernoulli and constant terms into the polynomial. $\square$

An equivalent expansion is Adamchik's Proposition 2 [1]; the spelling above isolates the one z -dependent transcendental.

7.2 Reflection

Theorem 7.2 (reflection at the jet floor). *For every integer $k \geq 0$ and $0 < z < 1$,*

$$\zeta'(-k, z) + (-1)^k \zeta'(-k, 1-z) = (-1)^{\lfloor k/2 \rfloor} \frac{k!}{(2\pi)^k} \text{Cl}_{k+1}(2\pi z),$$

where $\text{Cl}_{2m}(\theta) = \text{Im Li}_{2m}(e^{i\theta})$, $\text{Cl}_{2m+1}(\theta) = \text{Re Li}_{2m+1}(e^{i\theta})$.

Proof. In Hurwitz's formula, written as $\zeta(1-s, z) = \frac{2\Gamma(s)}{(2\pi)^s} [\cos(\frac{\pi s}{2})C_s(z) + \sin(\frac{\pi s}{2})S_s(z)]$ with C_s, S_s the cosine/sine Dirichlet series, the combination even (odd) under $z \mapsto 1-z$ keeps only the C (S) term. At the integer $s = k+1$, exactly one of $\cos \frac{\pi s}{2}$, $\sin \frac{\pi s}{2}$ vanishes, and it is the one attached to the parity-matching combination; differentiating in s there, the product rule retains only the term in which d/ds falls on the vanishing trigonometric factor, so a single *bare* Clausen value survives and no derivative of the Dirichlet series enters. Collecting constants gives the display. $\square$

For $k \geq 1$ this is the real part of Adamchik's Proposition 1 [1], stated there for every positive integer order with the Bernoulli term as the imaginary part (equivalently Miller–Adamchik [16], eq. (21)); the bare-Clausen real spelling, the $k = 0$ inclusion, and the mechanism note are the reformulation here. $k = 0$ is Kummer-flavoured ($\text{Cl}_1(2\pi z) = -\ln(2 \sin \pi z)$). The parity-*opposite* combination does not close: it produces the series $\sum \ln n \{\cos | \sin\}(2\pi n z)/n^{k+1}$ — the generalized-Glaisher atoms of §7.4.

Propagating Theorem 7.2 through the ladder yields reflection formulas at the $\psi^{(-n)}$ -level — a relation which Espinosa and Moll [12] record as not known to them (*"We have been unable to find a generalization of the other well-known functional relation for the polygamma function"*):

Theorem 7.3 (negapolygamma reflection).

$$\begin{aligned} \psi^{(-2)}(z) - \psi^{(-2)}(1-z) &= \frac{\text{Cl}_2(2\pi z)}{2\pi} + \left(z - \frac{1}{2}\right) \ln 2\pi, \\ \psi^{(-3)}(z) + \psi^{(-3)}(1-z) &= -\frac{\text{Cl}_3(2\pi z)}{4\pi^2} + \left(\frac{z^2}{2} - \frac{z}{2} + \frac{1}{4}\right) \ln 2\pi - \zeta'(-1) - \zeta'(-2) + \frac{1}{12}, \\ \psi^{(-4)}(z) - \psi^{(-4)}(1-z) &= -\frac{\text{Cl}_4(2\pi z)}{8\pi^3} + \left(\frac{z^3}{6} - \frac{z^2}{4} + \frac{z}{4} - \frac{1}{12}\right) \ln 2\pi \\ &\quad + \left(\frac{1}{2} - z\right) (\zeta'(-1) + \zeta'(-2)) + \frac{z}{12} - \frac{1}{24}, \\ \psi^{(-5)}(z) + \psi^{(-5)}(1-z) &= \frac{\text{Cl}_5(2\pi z)}{16\pi^4} + \left(\frac{z^4}{24} - \frac{z^3}{12} + \frac{z^2}{8} - \frac{z}{12} + \frac{1}{48}\right) \ln 2\pi \\ &\quad + \left(\frac{z-z^2}{2}\right) (\zeta'(-1) + \zeta'(-2)) \\ &\quad - \frac{\zeta'(-1)}{6} - \frac{\zeta'(-2)}{4} - \frac{\zeta'(-3)}{6} - \frac{\zeta'(-4)}{12} + \frac{z^2-z}{24} + \frac{49}{4320}. \end{aligned}$$

7.3 Multiplication

Differentiating the Hurwitz distribution relation at $s = -k$ gives, for all $k \geq 0$, $m \geq 2$,

$$\sum_{j=0}^{m-1} \zeta' \left(-k, z + \frac{j}{m} \right) = m^{-k} \left[\zeta'(-k, mz) - \ln m \frac{B_{k+1}(mz)}{k+1} \right] \quad (20)$$

($k = 0$ is Gauss's multiplication theorem for $\ln \Gamma$), and through the ladder,

$$\sum_{j=0}^{m-1} \psi^{(-n)}\left(z + \frac{j}{m}\right) = m^{1-n} \left[\psi^{(-n)}(mz) - \ln m \frac{B_n(mz)}{n!} \right] + \rho_{n,m}(z)$$

with elementary $\rho_{n,m}$; for $n = 2$,

$$\rho_{2,m}(z) = \frac{(m-1)(2z+1)}{4} \ln 2\pi + \frac{m^2-1}{12m} - \frac{m^2-1}{m} \zeta'(-1),$$

so only the *rational* part of the remainder is $(m^2-1)/(12m)$. The ψ -level law is, in the "balanced" convention of their negapolygamma [14], Theorem 3.2 of Espinosa–Moll [12]; the form above is its transcription to the Wolfram convention.

7.4 Bridges

At rational arguments the layers $\zeta'(-k, p/q)$ decompose along Dirichlet characters, and two laws organize the resulting constants.

Proposition 7.4 (trivial-zero bridges). *Let χ be primitive or principal, $k \geq 1$. If $\chi(-1) = (-1)^k$ (so that $L(-k, \chi) = 0$, since $B_{k+1}(\chi) \neq 0$ exactly when $\chi(-1) = (-1)^{k+1}$), then $L'(-k, \chi)$ is an elementary multiple of $L(k+1, \bar{\chi})$. For the principal character and χ_{-4} :*

$$\zeta'(-2k) = \frac{(-1)^k (2k)!}{2(2\pi)^{2k}} \zeta(2k+1), \quad \beta'(1-2k) = (-1)^{k+1} (2k-1)! \frac{2^{2k-1}}{\pi^{2k-1}} \beta(2k),$$

so in particular $\beta'(-1) = 2G/\pi$ (G Catalan's constant) and $\beta'(-3) = -48\beta(4)/\pi^3$.

Theorem 7.5 (harmonic-number bridge). *Let χ be a primitive character mod q with $\chi(-1) = (-1)^{k+1}$ (so that $L(-k, \chi) \neq 0$; for χ principal read ζ and require k odd). Then*

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \overline{\left(\frac{L'(-k, \chi)}{L(-k, \chi)} \right)} = \gamma + \ln \frac{2\pi}{q} - H_k.$$

For real χ the conjugation bar is invisible; for complex χ it is essential — equivalently, the bridge pairs χ at $s = k+1$ with $\bar{\chi}$ at $s = -k$, and the un-conjugated sum has the purely imaginary defect $2i \operatorname{Im}[L'(-k, \chi)/L(-k, \chi)]$.

Proof. Write the completed function $\Lambda(s) = (q/\pi)^{(s+\delta)/2} \Gamma\left(\frac{s+\delta}{2}\right) L(s, \chi)$ with $\delta = (1 - \chi(-1))/2$, so $\Lambda(s) = \varepsilon \bar{\Lambda}(1 - \bar{s})$, whose logarithmic derivative at real s gives $\Lambda'/\Lambda(s) = -\bar{\Lambda}'/\bar{\Lambda}(1 - \bar{s})$; at $s = k+1$,

$$\frac{L'(k+1, \chi)}{L(k+1, \chi)} + \overline{\left(\frac{L'(-k, \chi)}{L(-k, \chi)} \right)} = -\ln \frac{q}{\pi} - \frac{1}{2} \psi\left(\frac{k+1+\delta}{2}\right) - \frac{1}{2} \psi\left(\frac{\delta-k}{2}\right).$$

Since $\chi(-1) = (-1)^{k+1}$ means $\delta \equiv k+1 \pmod{2}$, the argument $(\delta-k)/2$ is a (possibly negative) half-integer, where $\cot \pi x = 0$; the reflection formula thus gives $\psi\left(\frac{\delta-k}{2}\right) = \psi\left(\frac{k+2-\delta}{2}\right)$, and $\left\{ \frac{k+1+\delta}{2}, \frac{k+2-\delta}{2} \right\}$ is a duplication pair $\{x, x + \frac{1}{2}\}$ with $2x = k+1$. Hence the two digammas sum to $2\psi(k+1) - 2\ln 2 = 2(H_k - \gamma) - 2\ln 2$, and the display follows. $\square$

Consequently, at each level $-k$ every character direction of the grid $\{\zeta'(-k, j/q)\}$ either reduces (trivial-zero bridge: to the $\zeta(k+1)$ /Catalan-family values) or carries exactly one new constant (the parity-opposite L'), the harmonic-number bridge converting freely between its $s = k+1$ and $s = -k$ spellings. Exact rational row-reduction of the full known-relation lattice (all reflection rows and all-divisor multiplication rows) makes this quantitative:

Proposition 7.6 (jet-grid residual dimension). *For every $3 \leq q \leq 30$ and $1 \leq k \leq 4$, the residual dimension of the grid $\{\zeta'(-k, j/q) : 1 \leq j \leq q-1\}$ modulo the lattice generated by the reflection and multiplication rows equals*

$$r_{q,k} = \#\{\chi \text{ primitive of conductor } f \mid q, f > 1, \chi(-1) = (-1)^{k+1}\} = \begin{cases} \varphi(q)/2 - 1, & k \text{ odd,} \\ \varphi(q)/2, & k \text{ even,} \end{cases}$$

with no exceptions.

The epistemic status deserves precision. For each fixed (q, k) in the stated range this *is* a theorem: the relation lattice is a finite, explicitly known matrix over $\mathbb{Q}$, and the rank computation is exact rational row reduction (no floating point; the script and its full 112-cell output accompany the article in the repository). The character-count *law*, as a statement for all q , is at present a verified conjecture, with the expected mechanism visible in the computation: decomposing the grid along Dirichlet characters, the reflection rows annihilate exactly the $(-1)^k$ -parity eigenspace (whose L -jets are the Clausen material already on the reflection right-hand side), the divisor multiplication rows pin the imprimitive and principal directions (the constant $\zeta'(-k)$ belongs to the right-hand-side inventory: a generalized Glaisher constant $\ln A_k$ for k odd, a multiple of $\zeta(k+1)$ for k even), and the surviving directions are precisely the parity-opposite primitive characters of Theorem 7.5. For k odd the count coincides with the residual law of the γ_1 -grids — the two lattices see the same even-character obstruction space.

7.5 Rational values

Theorem 7.7 (the half-argument tower). *For every $n \geq 2$,*

$$\psi^{(-n)}\left(\frac{1}{2}\right) = \frac{(2^{1-n} - 2) \zeta'(1-n) - 2^{1-n} \ln 2 \cdot B_n/n}{(n-1)!} + Q_n\left(\frac{1}{2}\right);$$

in lowest cases

$$\psi^{(-2)}\left(\frac{1}{2}\right) = \frac{3}{2} \ln A + \frac{5 \ln 2}{24} + \frac{\ln \pi}{4} \quad (\text{classical}), \quad \psi^{(-3)}\left(\frac{1}{2}\right) = \frac{7 \zeta(3)}{32\pi^2} + \frac{\ln A}{2} + \frac{\ln 2\pi}{16}.$$

Theorem 7.8 (third- and quarter-argument values at $n = 3, 4$).

$$\begin{aligned} \psi^{(-3)}\left(\frac{1}{4}\right) &= \frac{35 \zeta(3)}{256\pi^2} + \frac{\beta'(3)}{4\pi^3} + \frac{\ln A}{4} + \frac{\ln 2\pi}{128} - \frac{\gamma}{128}, \\ \psi^{(-3)}\left(\frac{1}{3}\right) &= \frac{13 \zeta(3)}{72\pi^2} + \frac{\sqrt{3} L'(3, \chi_{-3})}{8\pi^3} + \frac{\ln A}{3} + \frac{7 \ln 2\pi}{324} - \frac{\gamma}{162}, \end{aligned}$$

via the exact spellings $\beta'(-2) = \frac{1}{4}(3 - 2\gamma - 2 \ln \frac{\pi}{2}) + 16\beta'(3)/\pi^3$ and $L'(-2, \chi_{-3}) = \frac{2}{9}(\ln \frac{3}{2\pi} + \frac{3}{2} - \gamma) + \frac{9\sqrt{3}}{2\pi^3} L'(3, \chi_{-3})$; and at the next level the parity alternates, so that $\psi^{(-4)}(\frac{1}{4})$ is explicit with no new constant:

$$\psi^{(-4)}\left(\frac{1}{4}\right) = \frac{\zeta'(-3, \frac{1}{4}) - \zeta'(-3)}{6} + Q_4\left(\frac{1}{4}\right), \quad \zeta'(-3, \frac{1}{4}) = \frac{Z + D}{2}$$

with the even part $Z = (4^{-3} - 2^{-3})\zeta'(-3) + (4^{-3} \ln 4 - 2^{-3} \ln 2)\zeta(-3)$ and the odd part $D = -48\beta(4)/(64\pi^3)$ from Proposition 7.4.

Remark 7.9. $\psi^{(-3)}(\frac{1}{2})$ appears in Adamchik [1]; the flattened quarter- and third-values appear to be new — notably, Mathematica's `FunctionExpand` reduces `PolyGamma[-3, 1/2]` fully but leaves the jet $\zeta'(-2, \frac{1}{4})$ unevaluated inside `PolyGamma[-3, 1/4]`: the atom $\beta'(3)$ is exactly what is missing. The same atom resolves a naive integer-relation search over $\{\zeta(3)/\pi^2, \ln A, \log s\}$, which must fail for this value.

Corollary 7.10 (reflection and transport companions). *With the same atoms as Theorem 7.8,*

$$\begin{aligned}\psi^{(-3)}\left(\frac{2}{3}\right) &= \frac{13\zeta(3)}{72\pi^2} - \frac{\sqrt{3}L'(3, \chi_{-3})}{8\pi^3} + \frac{2\ln A}{3} + \frac{19\ln 2\pi}{162} + \frac{\gamma}{162}, \\ \psi^{(-3)}\left(\frac{3}{4}\right) &= \frac{35\zeta(3)}{256\pi^2} - \frac{\beta'(3)}{4\pi^3} + \frac{3\ln A}{4} + \frac{19\ln 2\pi}{128} + \frac{\gamma}{128}, \\ \psi^{(-3)}\left(\frac{1}{6}\right) &= \frac{\zeta(3)}{12\pi^2} + \frac{5\sqrt{3}L'(3, \chi_{-3})}{32\pi^3} + \frac{\ln A}{6} - \frac{\ln 2\pi}{1296} - \frac{\ln 2}{648} - \frac{5\gamma}{648}, \\ \psi^{(-3)}\left(\frac{5}{6}\right) &= \frac{\zeta(3)}{12\pi^2} - \frac{5\sqrt{3}L'(3, \chi_{-3})}{32\pi^3} + \frac{5\ln A}{6} + \frac{235\ln 2\pi}{1296} + \frac{\ln 2}{648} + \frac{5\gamma}{648}.\end{aligned}$$

Proof. The first two from Theorem 7.3 with $\text{Cl}_3(2\pi/3) = -\frac{4}{9}\zeta(3)$ and $\text{Cl}_3(\pi/2) = -\frac{3}{32}\zeta(3)$; the sixths from the transport $\zeta(s, \frac{1}{6}) - \zeta(s, \frac{5}{6}) = (6^s + 3^s)L(s, \chi_{-3})$ differentiated at $s = -2$, the even part being $\zeta'(-2, \frac{1}{6}) + \zeta'(-2, \frac{5}{6}) = \frac{2}{3}\zeta'(-2)$ by inclusion–exclusion. The entire sixth family thus rides on the one atom $L'(3, \chi_{-3})$. $\square$

7.6 Denominators 5, 8, 12: quadratic and quartic atoms

The denominators 5, 8, 12 are the first where Proposition 7.6 forces genuinely new constants beyond the χ_{-3}/χ_{-4} families: at $k = 1$ each grid carries one new atom (the even real quadratic direction), and at $k = 2$, $q = 5$ the residual space is two-dimensional (a conjugate quartic pair).

Theorem 7.11 (the $k = 1$ grids). *Write $Z_1 = \zeta'(-1)$ and let χ_q be the even real quadratic primitive character mod $q \in \{5, 8, 12\}$ (Legendre mod 5; Kronecker $(8|\cdot)$, $(12|\cdot)$), with the exact values $L(-1, \chi_5) = -\frac{2}{5}$, $L(-1, \chi_8) = -1$, $L(-1, \chi_{12}) = -2$. Then for $\gcd(p, q) = 1$,*

$$\begin{aligned}\zeta'(-1, \frac{p}{5}) &= -\frac{Z_1}{5} + \frac{\text{Cl}_2(2\pi p/5)}{4\pi} + \chi_5(p) \left[\frac{L'(-1, \chi_5)}{20} - \frac{\ln 5}{50} \right] - \frac{\ln 5}{240}, \\ \zeta'(-1, \frac{p}{8}) &= -\frac{Z_1}{32} + \frac{\text{Cl}_2(\pi p/4)}{4\pi} + \chi_8(p) \left[\frac{L'(-1, \chi_8)}{32} - \frac{3\ln 2}{32} \right] + \frac{\ln 2}{384}, \\ \zeta'(-1, \frac{p}{12}) &= \frac{Z_1}{24} + \frac{\text{Cl}_2(\pi p/6)}{4\pi} + \chi_{12}(p) \left[\frac{L'(-1, \chi_{12})}{48} - \frac{\ln 2}{12} - \frac{\ln 3}{24} \right] + \frac{\ln 3}{576},\end{aligned}$$

whence $\psi^{(-2)}(p/q) = \zeta'(-1, p/q) - Z_1 + Q_2(p/q)$ through the ladder; for instance

$$\psi^{(-2)}\left(\frac{1}{5}\right) = -\frac{6Z_1}{5} + \frac{\text{Cl}_2(2\pi/5)}{4\pi} + \frac{L'(-1, \chi_5)}{20} - \frac{29\ln 5}{1200} + \frac{\ln 2\pi}{10} + \frac{2}{25}.$$

Proof. On each grid the two reflection rows (Theorem 7.2), the multiplication rows (20) for every divisor (non-coprime entries reduced through the sub-grids $q \in \{2, 3, 4, 6\}$), and the character row $\sum_p \chi_q(p) \zeta'(-1, \frac{p}{q}) = \frac{1}{q} [L'(-1, \chi_q) + \ln q L(-1, \chi_q)]$ form a system of full rank

$\varphi(q) = 4$; solving it and reducing the composite-argument Clausen values ($\text{Cl}_2(\frac{3\pi}{4}) = \text{Cl}_2(\frac{\pi}{4}) - \frac{G}{2}$, $\text{Cl}_2(\frac{\pi}{6}) = \frac{2}{3}G + \frac{3}{8}\text{Cl}_2(\frac{2\pi}{3})$) gives the display. For the composite q the known rows are one short of independent — the single dependency is itself a Clausen duplication law in disguise (at $q = 8$: $\text{refl}(\frac{1}{8}) - \text{refl}(\frac{3}{8}) = m_2(\frac{1}{8}) - m_2(\frac{3}{8})$ is equivalent to $\text{Cl}_2(\frac{3\pi}{4}) = \text{Cl}_2(\frac{\pi}{4}) - \frac{G}{2}$) — and the quadratic character row supplies the missing dimension, exactly as Proposition 7.6 predicts ($r_{q,1} = 1$ for $q \in \{5, 8, 12\}$). $\square$

Theorem 7.12 (the quartic floor: $k = 2, q = 5$). *Let χ be the odd quartic character mod 5 with $\chi(2) = i$, for which $L(-2, \chi) = -\frac{4+2i}{5}$ exactly, and write $L'(-2, \chi) = R + iI$. Then*

$$\begin{aligned}\zeta'(-2, \tfrac{1}{5}), \zeta'(-2, \tfrac{4}{5}) &= -\frac{\text{Cl}_3(2\pi/5)}{4\pi^2} \pm \left[\frac{R}{50} - \frac{2\ln 5}{125} \right], \\ \zeta'(-2, \tfrac{2}{5}), \zeta'(-2, \tfrac{3}{5}) &= -\frac{\text{Cl}_3(4\pi/5)}{4\pi^2} \pm \left[\frac{I}{50} - \frac{\ln 5}{125} \right],\end{aligned}$$

and, flattening through the ladder with $\text{Cl}_3(\frac{2\pi}{5}), \text{Cl}_3(\frac{4\pi}{5}) = -\frac{6}{25}\zeta(3) \pm \frac{\sqrt{5}}{4}L(3, \chi_5)$,

$$\begin{aligned}\psi^{(-3)}(\tfrac{1}{5}) &= \frac{31\zeta(3)}{200\pi^2} - \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} + \frac{R}{100} - \frac{\ln 5}{125} + \frac{\ln 2\pi}{100} + \frac{7}{1500} - \frac{Z_1}{5}, \\ \psi^{(-3)}(\tfrac{2}{5}) &= \frac{31\zeta(3)}{200\pi^2} + \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} + \frac{I}{100} - \frac{\ln 5}{250} + \frac{\ln 2\pi}{25} + \frac{41}{1500} - \frac{2Z_1}{5}, \\ \psi^{(-3)}(\tfrac{3}{5}) &= \frac{31\zeta(3)}{200\pi^2} + \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} - \frac{I}{100} + \frac{\ln 5}{250} + \frac{9\ln 2\pi}{100} + \frac{7}{125} - \frac{3Z_1}{5}, \\ \psi^{(-3)}(\tfrac{4}{5}) &= \frac{31\zeta(3)}{200\pi^2} - \frac{\sqrt{5}L(3, \chi_5)}{32\pi^2} - \frac{R}{100} + \frac{\ln 5}{125} + \frac{4\ln 2\pi}{25} + \frac{59}{750} - \frac{4Z_1}{5}.\end{aligned}$$

Remark 7.13. The genuinely new constants here are exactly the quartic pair $\{R, I\}$, matching $r_{5,2} = \varphi(5)/2 = 2$ of Proposition 7.6; the multiplication row $\sum_p \zeta'(-2, \frac{p}{5}) = \frac{6}{25}\zeta(3)/\pi^2$ is dependent on the reflections — it is the Clausen multiplication law $\text{Cl}_3(\frac{2\pi}{5}) + \text{Cl}_3(\frac{4\pi}{5}) = -\frac{12}{25}\zeta(3)$ in disguise. Through the conjugated harmonic-number bridge (Theorem 7.5 with $k = 2, H_2 = \frac{3}{2}$) the pair $\{R, I\}$ is interchangeable with $\{\text{Re}, \text{Im}\}$ of $L'(3, \chi)$.

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Appendix: fully expanded values at the third and quarter points

The reduction recipes of Theorems 6.4 and 6.6, carried out to the end, give the following flattened closed forms. The Γ_2 values live in the atom set announced in Theorem 6.4; the Γ_3 values pick up, exactly as Theorem 6.6 predicts, the three further atoms γ_3, π^4 and $\gamma_4(\frac{1}{q}) - \gamma_4(1 - \frac{1}{q})$. The combinations $\gamma + \ln 6\pi$ and $\gamma + \ln 8\pi$ are the constants Λ_3, Λ_4 of Theorem 4.1. Each

display has been verified at 90-digit working precision (residuals below 10^{-88}) against $\Gamma_n(p) = \frac{(-1)^{n+1}}{n+1} [\zeta^{(n+1)}(0, p) - \zeta^{(n+1)}(0)]$ evaluated from Hurwitz-zeta derivative jets.

$$\begin{aligned}
\Gamma_2\left(\frac{1}{3}\right) &= \frac{1}{2} \gamma_2 + \gamma \gamma_1 + \gamma_1 \ln 2\pi - \frac{1}{2} \gamma_1 \ln 3 \\
&+ \frac{4}{9} \gamma^3 + \frac{4}{3} \gamma^2 \ln 2\pi - \frac{1}{6} \gamma^2 \ln 3 + \frac{4}{3} \gamma \ln^2 2\pi \\
&+ \frac{7}{6} \gamma \ln 2\pi \ln 3 - \frac{1}{6} \gamma \ln^2 3 - \frac{4}{9} \zeta(3) - \frac{\pi^2}{12} \ln 2\pi \\
&+ \frac{\pi^2}{24} \ln 3 + \frac{4}{9} \ln^3 2\pi + \frac{4}{3} \ln^2 2\pi \ln 3 + \frac{7}{12} \ln 2\pi \ln^2 3 \\
&- \frac{1}{18} \ln^3 3 \\
&- \left((\gamma + \ln 6\pi)^2 - \frac{\pi^2}{12} \right) \ln \Gamma\left(\frac{1}{3}\right) \\
&- \frac{\sqrt{3}}{6\pi} (\gamma + \ln 6\pi) (\gamma_2(\frac{1}{3}) - \gamma_2(\frac{2}{3})) \\
&- \frac{\sqrt{3}}{18\pi} (\gamma_3(\frac{1}{3}) - \gamma_3(\frac{2}{3}))
\end{aligned}$$

$$\begin{aligned}
\Gamma_2\left(\frac{1}{4}\right) &= \frac{1}{2} \gamma_2 + \gamma \gamma_1 + \frac{1}{2} \gamma_1 \ln 2 + \gamma_1 \ln \pi \\
&+ \frac{1}{2} \gamma^3 + 2 \gamma^2 \ln 2 + \frac{3}{2} \gamma^2 \ln \pi + 6 \gamma \ln^2 2 \\
&+ \frac{13}{2} \gamma \ln 2 \ln \pi + \frac{3}{2} \gamma \ln^2 \pi - \frac{1}{2} \zeta(3) - \frac{\pi^2}{24} \ln 2 \\
&- \frac{\pi^2}{12} \ln \pi + \frac{49}{6} \ln^3 2 + \frac{45}{4} \ln^2 2 \ln \pi + \frac{9}{2} \ln 2 \ln^2 \pi \\
&+ \frac{1}{2} \ln^3 \pi \\
&- \left((\gamma + \ln 8\pi)^2 - \frac{\pi^2}{12} \right) \ln \Gamma\left(\frac{1}{4}\right) \\
&- \frac{1}{4\pi} (\gamma + \ln 8\pi) (\gamma_2(\frac{1}{4}) - \gamma_2(\frac{3}{4})) \\
&- \frac{1}{12\pi} (\gamma_3(\frac{1}{4}) - \gamma_3(\frac{3}{4}))
\end{aligned}$$

$$\begin{aligned}
\Gamma_3\left(\frac{1}{3}\right) = & -\frac{1}{2}\gamma_3 - \frac{3}{2}\gamma\gamma_2 - \frac{3}{2}\gamma_2\ln 2\pi + \frac{3}{4}\gamma_2\ln 3 \\
& - \frac{3}{2}\gamma^2\gamma_1 - 3\gamma\gamma_1\ln 2\pi + \frac{3}{2}\gamma\gamma_1\ln 3 - \frac{3}{2}\gamma_1\ln^2 2\pi \\
& + \frac{3}{2}\gamma_1\ln 2\pi\ln 3 + \frac{3}{4}\gamma_1\ln^2 3 - \frac{\pi^2}{8}\gamma_1 - \frac{1}{2}\gamma^4 \\
& - 2\gamma^3\ln 2\pi + \frac{1}{4}\gamma^3\ln 3 - 3\gamma^2\ln^2 2\pi - \frac{3}{2}\gamma^2\ln 2\pi\ln 3 \\
& + \frac{3}{8}\gamma^2\ln^2 3 - \frac{\pi^2}{24}\gamma^2 - 2\gamma\ln^3 2\pi - \frac{15}{4}\gamma\ln^2 2\pi\ln 3 \\
& - \frac{3}{2}\gamma\ln 2\pi\ln^2 3 + \frac{1}{4}\gamma\ln^3 3 + \frac{\pi^2}{6}\gamma\ln 2\pi - \frac{\pi^2}{48}\gamma\ln 3 \\
& + \frac{5\pi^2}{24}\ln^2 2\pi + \frac{\pi^2}{24}\ln 2\pi\ln 3 - \frac{7\pi^2}{96}\ln^2 3 + \frac{\pi^4}{120} \\
& - \frac{1}{2}\ln^4 2\pi - 2\ln^3 2\pi\ln 3 - \frac{15}{8}\ln^2 2\pi\ln^2 3 - \frac{1}{2}\ln 2\pi\ln^3 3 \\
& + \frac{1}{16}\ln^4 3 \\
& + \left((\gamma + \ln 6\pi)^3 - \frac{\pi^2}{4}(\gamma + \ln 6\pi) + 2\zeta(3) \right) \ln \Gamma\left(\frac{1}{3}\right) \\
& + \frac{\sqrt{3}}{48\pi} \left(12(\gamma + \ln 6\pi)^2 - \pi^2 \right) \left(\gamma_2\left(\frac{1}{3}\right) - \gamma_2\left(\frac{2}{3}\right) \right) \\
& + \frac{\sqrt{3}}{6\pi} (\gamma + \ln 6\pi) \left(\gamma_3\left(\frac{1}{3}\right) - \gamma_3\left(\frac{2}{3}\right) \right) \\
& + \frac{\sqrt{3}}{24\pi} \left(\gamma_4\left(\frac{1}{3}\right) - \gamma_4\left(\frac{2}{3}\right) \right)
\end{aligned}$$

$$\begin{aligned}
\Gamma_3\left(\frac{1}{4}\right) = & -\frac{1}{2}\gamma_3 - \frac{3}{2}\gamma\gamma_2 - \frac{3}{4}\gamma_2\ln 2 - \frac{3}{2}\gamma_2\ln\pi \\
& - \frac{3}{2}\gamma^2\gamma_1 - \frac{3}{2}\gamma\gamma_1\ln 2 - 3\gamma\gamma_1\ln\pi + \frac{9}{4}\gamma_1\ln^2 2 \\
& - \frac{3}{2}\gamma_1\ln 2\ln\pi - \frac{3}{2}\gamma_1\ln^2\pi - \frac{\pi^2}{8}\gamma_1 - \frac{9}{16}\gamma^4 \\
& - 3\gamma^3\ln 2 - \frac{9}{4}\gamma^3\ln\pi - \frac{45}{4}\gamma^2\ln^2 2 - \frac{51}{4}\gamma^2\ln 2\ln\pi \\
& - \frac{27}{8}\gamma^2\ln^2\pi - \frac{\pi^2}{32}\gamma^2 - 27\gamma\ln^3 2 - \frac{153}{4}\gamma\ln^2 2\ln\pi \\
& - \frac{33}{2}\gamma\ln 2\ln^2\pi - \frac{9}{4}\gamma\ln^3\pi + \frac{\pi^2}{4}\gamma\ln 2 + \frac{3\pi^2}{16}\gamma\ln\pi \\
& + \frac{3\pi^2}{8}\ln^2 2 + \frac{11\pi^2}{16}\ln 2\ln\pi + \frac{7\pi^2}{32}\ln^2\pi + \frac{29\pi^4}{3840} \\
& - \frac{207}{8}\ln^4 2 - \frac{179}{4}\ln^3 2\ln\pi - 27\ln^2 2\ln^2\pi - \frac{27}{4}\ln 2\ln^3\pi \\
& - \frac{9}{16}\ln^4\pi \\
& + \left((\gamma + \ln 8\pi)^3 - \frac{\pi^2}{4}(\gamma + \ln 8\pi) + 2\zeta(3)\right)\ln\Gamma\left(\frac{1}{4}\right) \\
& + \frac{1}{32\pi}\left(12(\gamma + \ln 8\pi)^2 - \pi^2\right)(\gamma_2(\tfrac{1}{4}) - \gamma_2(\tfrac{3}{4})) \\
& + \frac{1}{4\pi}(\gamma + \ln 8\pi)(\gamma_3(\tfrac{1}{4}) - \gamma_3(\tfrac{3}{4})) \\
& + \frac{1}{16\pi}(\gamma_4(\tfrac{1}{4}) - \gamma_4(\tfrac{3}{4}))
\end{aligned}$$

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